

PostgreSQL with Sequelize ORM

✓ Project: Node.js + PostgreSQL + Sequelize

📁 Folder Structure

pgsql

CopyEdit

project-root/

```
|
|
| └─ config/
|   └─ db.js
|
|
| └─ models/
|   └─ user.model.js
|   └─ blog.model.js
|   └─ index.js
|
|
| └─ controllers/
|   └─ user.controller.js
|   └─ blog.controller.js
|
|
| └─ routes/
|   └─ user.routes.js
|   └─ blog.routes.js
|
|
| └─ app.js
└─ package.json
```

1 Install Dependencies

bash

CopyEdit

npm init -y

npm install express sequelize pg pg-hstore

2 config/db.js (Sequelize Connection)

js

CopyEdit

```
const { Sequelize } = require('sequelize');
```

```
const sequelize = new Sequelize('node_app', 'postgres', 'your_password', {  
  host: 'localhost',  
  dialect: 'postgres',  
});
```

```
module.exports = sequelize;
```

3 models/user.model.js

js

CopyEdit

```
const { DataTypes } = require('sequelize');
```

```
const sequelize = require('../config/db');
```

```
const User = sequelize.define('User', {  
  email: {  
    type: DataTypes.STRING,  
    allowNull: false,  
    unique: true,  
  },
```

```
password: {  
  type: DataTypes.STRING,  
  allowNull: false,  
},  
});
```

```
module.exports = User;
```

4 models/blog.model.js

js

CopyEdit

```
const { DataTypes } = require('sequelize');  
const sequelize = require('../config/db');
```

```
const Blog = sequelize.define('Blog', {  
  title: {  
    type: DataTypes.STRING,  
    allowNull: false,  
  },  
  content: {  
    type: DataTypes.TEXT,  
  },  
});
```

```
module.exports = Blog;
```

5 models/index.js (Associations)

js

CopyEdit

```
const sequelize = require('../config/db');
const User = require('../user.model');
const Blog = require('../blog.model');

// Associations
User.hasMany(Blog, { foreignKey: 'authorId' });
Blog.belongsTo(User, { foreignKey: 'authorId' });

module.exports = { sequelize, User, Blog };
```

6 controllers/user.controller.js

js

CopyEdit

```
const { User } = require('../models');

exports.createUser = async (req, res) => {
  const { email, password } = req.body;
  const user = await User.create({ email, password });
  res.json(user);
};

exports getUsers = async (req, res) => {
  const users = await User.findAll();
  res.json(users);
};
```

7 controllers/blog.controller.js

js

CopyEdit

```
const { Blog, User } = require('../models');

exports.createBlog = async (req, res) => {
  const { title, content, authorId } = req.body;
  const blog = await Blog.create({ title, content, authorId });
  res.json(blog);
};

exports.getBlogs = async (req, res) => {
  const blogs = await Blog.findAll({ include: User });
  res.json(blogs);
};
```

8 routes/user.routes.js

js

CopyEdit

```
const express = require('express');
const router = express.Router();
const userController = require('../controllers/user.controller');
```

```
router.post('/', userController.createUser);
router.get('/', userController.getUsers);
```

```
module.exports = router;
```

9 routes/blog.routes.js

js

CopyEdit

```
const express = require('express');
```

```
const router = express.Router();

const blogController = require('../controllers/blog.controller');

router.post('/', blogController.createBlog);
router.get('/', blogController.getBlogs);

module.exports = router;
```

app.js

js

CopyEdit

```
const express = require('express');
const { sequelize } = require('./models');

const app = express();
app.use(express.json());

// Routes
app.use('/users', require('./routes/user.routes'));
app.use('/blogs', require('./routes/blog.routes'));

// DB Sync & Start
sequelize.sync().then(() => {
  console.log('DB Synced');
  app.listen(3000, () => console.log('Server running on port 3000'));
});
```

PostgreSQL Setup

sql

CopyEdit

-- Inside psql

CREATE DATABASE node_app;

\c node_app;

 To Run

bash

CopyEdit

node app.js

 Sample Requests (Postman or cURL)

Create User

json

CopyEdit

POST /users

```
{  
  "email": "test@example.com",  
  "password": "123456"  
}
```

Create Blog

json

CopyEdit

POST /blogs

```
{  
  "title": "First Blog",  
  "content": "Postgres with Sequelize!",  
  "authorId": 1  
}
```